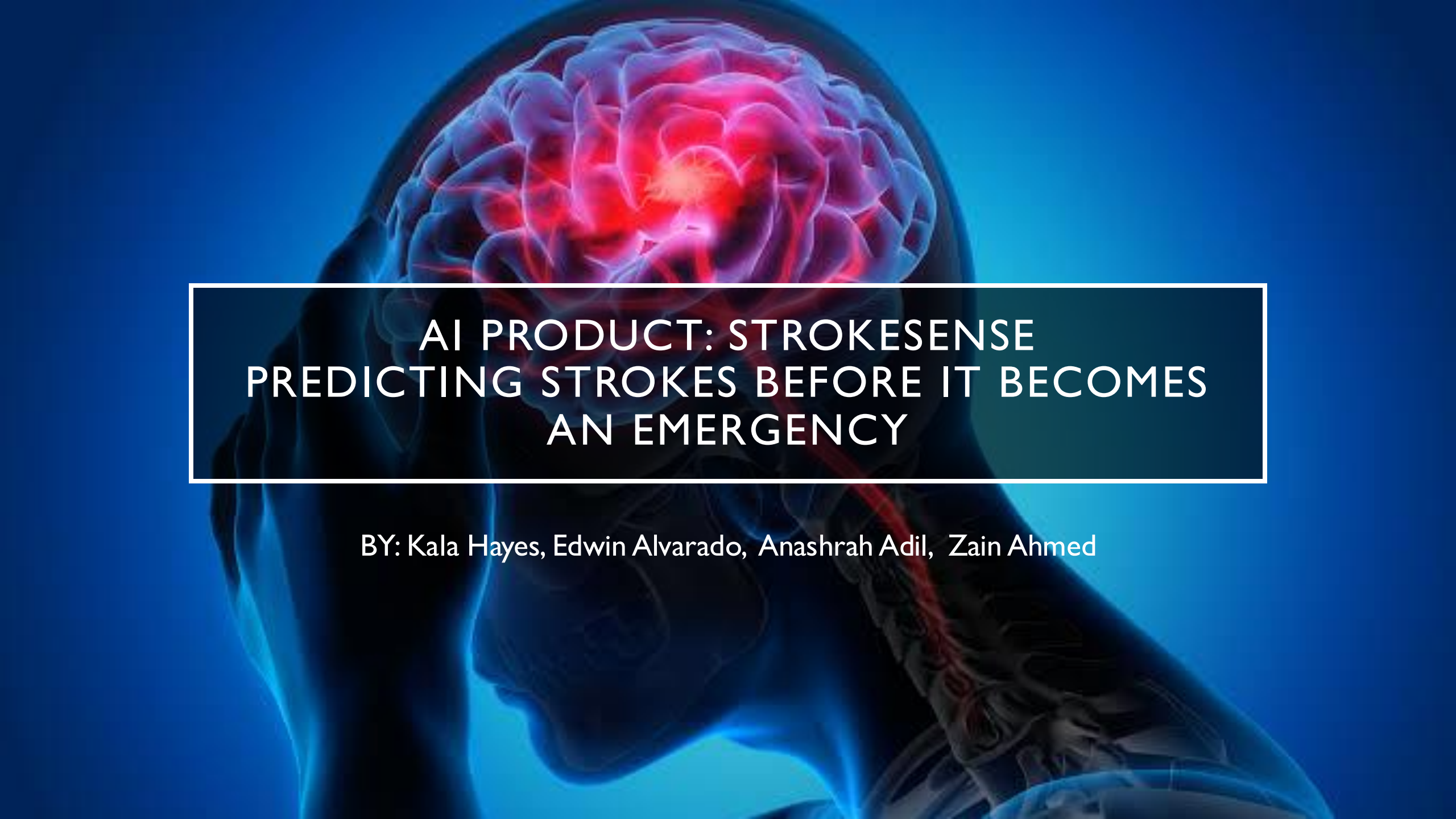




AI PRODUCT: STROKESENSE PREDICTING STROKES BEFORE IT BECOMES AN EMERGENCY

BY: Kala Hayes, Edwin Alvarado, Anashrah Adil, Zain Ahmed

PRODUCT OVERVIEW

What Is StrokeSense AI?

- A wearable AI health monitoring system designed to predict stroke risk.
- Uses machine learning to analyze continuous health data and identify risk patterns.
- Provides personalized stroke risk based on individual health information.

How it works

- Collects real-time data from wearable sensors.
- Combines wearable data with medical history and health information.
- Uses ML algorithms to identify changes associated with increased stroke risk
- Send alerts to users and healthcare providers when risk increases

Potential Users

- Adults with stroke risk factors
- Patient with hypertension, diabetes, or heart diseases
- Healthcare providers
- Caregivers supporting high-risk individuals

PROBLEM & SOLUTION

The Problem

- Stroke is a leading cause of death and disability
- Many patient have risk factor but show no symptoms.
- Early detection can reduce complications

Our Solution

StrokeSense AI

- A wearable AI device that continuously monitors health data and predicts stroke risk in real time.
- Supports physician with early intervention.
- Uses machine learning to identify high-risk patients.

MACHINE LEARNING PIPELINE

How StrokeSense AI Makes Predictions

1. Collect Data- Wearable gathers heart rate, blood pressure, activity, sleep, and ECG
2. Preprocess Data- Clean missing values, removes errors, and normalize data.
3. Feature Engineering: - Select key health indicators that influence stroke risk.
4. Train & Test Models- Compare Logistic Regression, Random Forest, Gradient Boosting, and SVM using historical patient data.
5. Evaluate Performance-Choose the best model based on Precision, Recall, F1-score, and ROC-AUC.
6. Predict stroke risk - Generate a personalized risk score and send alerts to the users and healthcare provider if needed.

DATA SOURCES

Data used for Prediction

- Heart Rate
- Blood pressure
- Blood oxygen
- Activity levels
- Sleep Patterns
- ECG (if supported by the wearable)
- Historical Electronic Health Records
- Medical History (hypertension, diabetes, smoking)

(KALA HAYES, EDWIN ALVARADO, ANASHRAH ADIL, ZAIN AHMED)-AI ETHICIST

Ethical Considerations

- The AI must prevent bias work equally well for people of different ages, races, accents, disabilities, and medical conditions.
- Privacy
- The robot collects highly sensitive health information. Patients must control who accesses their data.
- Transparency
- Users should understand why the AI made a recommendation, such as calling or sending an alert.
- Accountability
- If the AI makes an incorrect decision, responsibility should be shared among the developers, healthcare providers, and the company maintaining the system.

EDWIN ALVARADO – DATA SCIENTIST

Building a Fair Model

- Train on diverse patient data.
- Remove missing and duplicate data.
- Compare multiple ML algorithms.
- Evaluated using Precision, Recall, F1-score, and ROC- AUC.
- Continually monitor for bias and accuracy.

KALA HAYES –PRODUCT MANAGER

Bringing StrokeSense AI to Market

- Integrate with a smartwatch or fitness wearable.
- Monitor health continuously in real time.
- Send early stroke risk alert to users and healthcare providers.
- Improve preventive care through ongoing monitoring
- Increase users' trust with secure, transparent data practices.

ZAIN AHMED: CONSUMER ADVOCATE

Protecting the User

- Users controls what health data is shared
- Personal data is encrypted and protected
- AI predictions should be easy to understand
- Users can opt out of data collection at anytime
- AI should support- not replace – a doctor's judgment.

ANASHRAH ADIL: GOVERNMENT REGULATOR

Safe & Responsible AI

- Requires HIPPA-compliant data protection.
- Conduct regular Bias and Fairness audits.
- Validate the AI before public release
- Monitor model performance after deployment
- Require human oversight for high-risk alerts.

SOCIETAL IMPACT

Benefits

- Detects stroke risk earlier
- Encourages prevention healthcare.
- Supports physicians with real-time data.
- May reduce stroke-related deaths and disability.

Challenges

- Protecting sensitive health information
- Preventing Bias across different population.
- Avoiding False alarms or missed predications
- Ensuring affordable access for all users.

PANEL DISCUSSION

Questions for Discussion

- Should AI send emergency alerts automatically?
- Who is responsible if the AI makes a wrong predication?
- How can developers reduce bias in healthcare AI?
- Should insurance companies access AI risk scores?
- How much should people rely on AI for health decisions?

TEAM'S POSITION

Team answers for Discussion

- Should AI send emergency alerts automatically?
 - Yes, but with safeguards.
- Who is responsible if the AI makes a wrong predication?
 - Responsibility should be shared based on what caused the error.
- How can developers reduce bias in healthcare AI?
 - Use diverse data and continuously test the model.
- Should insurance companies access AI risk scores?
 - No, not for determining coverage, premiums, or eligibility
- How much should people rely on AI for health decisions?
 - AI should assist people, not replace medical professionals.

CONCLUSION

Key Takeaways

- StrokeSense AI Uses supervised machine learning to predict stroke risk.
- Continuous wearable data improves prediction accuracy
- Ethical AI requires privacy, fairness, transparency, and accountability.
- Ai should assist health care provider- not replace them.